UNIT 4

Multi-parameter Stress Field, Mixed Modes, SIFs & Plastic Zone Models

Williams series | T-stress | Modes II & III | mixed mode | SIF handbook | Irwin | Dugdale

1. Multi-parameter Stress Field (Mode I)

The Westergaard/Irwin field $\sigma_{ij} = K/\sqrt{(2\pi r)}\ f_{ij}(\theta)$ is only the **first term** of an infinite series. Near the tip ($r \to 0$) it dominates, but at finite r (small specimens, short cracks, experiments) higher-order terms matter.

1.1 Williams eigenfunction expansion (1957)

Solve $\nabla^4\varphi = 0$ for a wedge of 2π with traction-free flanks, $\varphi = r^{n/2+1}\ F_n(\theta)$. For Mode I (symmetric):

$$\sigma_x = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [2 + n/2 + (-1)^n]\ \cos(n/2-1)\theta - (n/2-1)\ \cos(n/2-3)\theta\ \}$$

$$\sigma_y = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [2 - n/2 - (-1)^n]\ \cos(n/2-1)\theta + (n/2-1)\ \cos(n/2-3)\theta\ \}$$

$$\tau_{xy} = \Sigma_n\ A_n\ (n/2)\ r^{n/2-1}\ \{\ [n/2 + (-1)^n]\ \sin(n/2-1)\theta - (n/2-1)\ \sin(n/2-3)\theta\ \}$$

Term n	Order in r	Meaning
1	$r^{-1/2}$	singular term: $A_1 = K_I/\sqrt{(2\pi)}$ → K-field
2	r^0	uniform stress parallel to crack: $\sigma_{x0} = T = 4A_2$ (**T-stress**)
3	$r^{1/2}$	first non-singular higher-order term, A_3 …
≥ 4	r^1, $r^{3/2}$ …	further terms; matter far from tip

Why multi-parameter fields?

- Single-term (K) data fit only for $r \to 0$, but near the tip plasticity, 3-D effects and measurement noise spoil data; far data need more terms.
- Photoelastic / strain-gauge / DIC measurements are taken at r = 0.3–1 × thickness: using 3–5 terms with over-deterministic **least-squares** (Sanford & Dally; Ramesh) gives accurate K and T.
- Short cracks, edge cracks and specimen geometry change the higher terms ⇒ explain size/geometry (**constraint**) effects on toughness.

1.2 T-stress and constraint

Three-term field: $\sigma_{ij} = K_I/\sqrt{(2\pi r)}\, f_{ij}(\theta) + T\,\delta_{1i}\delta_{1j} + O(r^{1/2})$. T acts parallel to the crack plane (σ_x).

$$\text{Biaxiality ratio } B = T\sqrt{(\pi a)} / K_I$$

- Centre crack, uniaxial $\sigma \perp$ crack: $T = -\sigma$ ($B = -1$). Equi-biaxial loading: $T = 0$. Edge-cracked bend (SENB) deep crack: $T \approx -0.4\sigma$... (negative, **low constraint**). Deep CT: T slightly positive (high constraint).
- **$T > 0$** (high constraint): plastic zone elongates forward, high triaxiality, lower toughness; **$T < 0$** (low constraint): zone rotates/enlarges, triaxiality lost, toughness higher.
- T-stress also governs **directional stability** of crack path: $T > 0$ → path unstable (deviates), $T < 0$ → stable.
- Two-parameter approaches: **K–T** (elastic) and **J–Q** (O'Dowd & Shih; Q = difference from SSY field). They explain why small specimens (low constraint) give higher apparent toughness than structures.

2. Mode II, Mode III and Mixed Modes

2.1 Mode II (in-plane shear) near-tip field

$$\sigma_x = -K_{II}/\sqrt{(2\pi r)} \cdot \sin(\theta/2)\,[\,2 + \cos(\theta/2)\cos(3\theta/2)\,]$$

$$\sigma_y = K_{II}/\sqrt{(2\pi r)} \cdot \sin(\theta/2)\cos(\theta/2)\cos(3\theta/2)$$

$$\tau_{xy} = K_{II}/\sqrt{(2\pi r)} \cdot \cos(\theta/2)\,[\,1 - \sin(\theta/2)\sin(3\theta/2)\,]$$

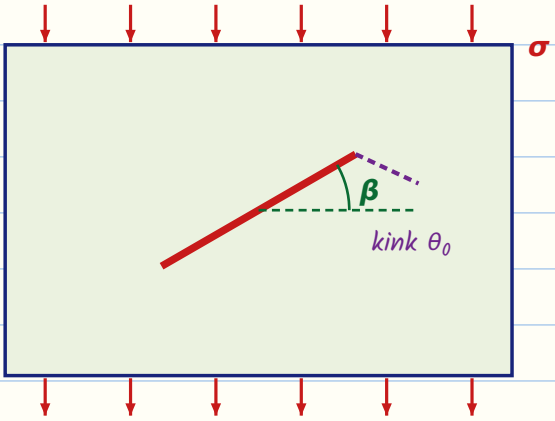
- $K_{II} = \tau\sqrt{(\pi a)}$ (centre crack, remote shear τ). Displacements: $u = (K_{II}/2\mu)\sqrt{(r/2\pi)}\,\sin(\theta/2)[\kappa + 1 + 2\cos^2(\theta/2)]$, $v = -(K_{II}/2\mu)\sqrt{(r/2\pi)}\,\cos(\theta/2)[\kappa - 1 - 2\sin^2(\theta/2)]$.
- On crack line $\theta = 0$: $\sigma_x = \sigma_y = 0$ and $\tau_{xy} = K_{II}/\sqrt{(2\pi r)}$ — pure shear ahead of tip. σ_x is antisymmetric ⇒ maximum tension is **off-axis**.

2.2 Mode III (anti-plane shear)

$$\tau_{xz} = -K_{III}/\sqrt{(2\pi r)}\,\sin(\theta/2)\ ;\ \tau_{yz} = K_{III}/\sqrt{(2\pi r)}\,\cos(\theta/2)\ ;\ w = (2K_{III}/\mu)\sqrt{(r/2\pi)}\,\sin(\theta/2)$$

- $K_{III} = \tau\sqrt{(\pi a)}$. Only one displacement $w(x, y)$ ⇒ simplest: governed by Laplace equation $\nabla^2 w = 0$. $G_{III} = K_{III}^2/(2\mu) = (1+\nu)K_{III}^2/E$.

2.3 Mixed-mode loading



Inclined crack (β from ⊥ to σ):

$\sigma_n = \sigma \cos^2\beta$, $\tau = \sigma \sin\beta \cos\beta$

$K_I = \sigma\sqrt{(\pi a)}\cos^2\beta$

$K_{II} = \sigma\sqrt{(\pi a)}\sin\beta\cos\beta$

$\beta = 0$: pure Mode I
$\beta = 45°$: $K_I = 0.5K$, $K_{II} = 0.5K$
$\beta = 90°$: crack ∥ load, $K = 0$

Fig 4.1 Inclined crack – mixed Mode I + II

- Superposition: $\sigma_{ij} = [K_I f^{\wedge}I_{ij} + K_{II} f^{\wedge}II_{ij} + K_{III} f^{\wedge}III_{ij}]/\sqrt{(2\pi r)}$. Energy: $G = (K_I^2 + K_{II}^2)/E' + K_{III}^2/2\mu$.
- Mode-mixity angle $\psi = \tan^{-1}(K_{II}/K_I)$. Real structures are rarely pure Mode I (inclined cracks, welds, shafts under torsion + bending).
- Under mixed I + II the crack **kinks** and grows at an angle θ_0 to its original plane; the criteria (MTS, SED, G_{max}) are discussed in Unit V.

Worked Example 1 (inclined crack)

Central crack $2a = 20$ mm inclined $\beta = 30°$ to the plane ⊥ load, $\sigma = 100$ MPa (infinite plate, $E' = 210$ GPa). $K_0 = \sigma\sqrt{(\pi a)} = 100\sqrt{(\pi \times 0.01)} = 17.72$ MPa√m.

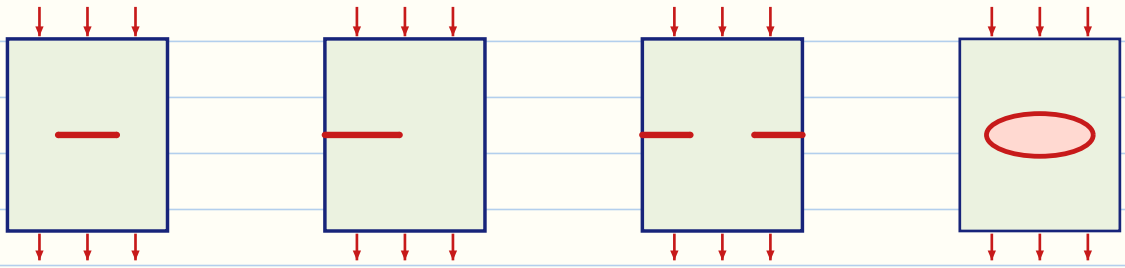
$K_I = K_0 \cos^2 30° =$ **13.29 MPa√m** ; $K_{II} = K_0 \sin 30° \cos 30° =$ **7.67 MPa√m** ; $G = (K_I^2 + K_{II}^2)/E' = (13.29^2 + 7.67^2)\times10^{12}/(210\times10^9) =$ **1.12 kJ/m²**.

MTS kink angle: $\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/(4K_{II})] = 2\tan^{-1}[(13.29 - 25.44)/30.70]$ = **−43.2°** (crack turns about 43° from its plane).

3. Stress Intensity Factors for Various Geometries

$$K_I = Y \sigma \sqrt{(\pi a)}$$

Y = geometry (correction) factor



Centre crack (CCT)	Single edge (SENT)	Double edge (DENT)	Embedded penny
$Y = \sqrt{\sec(\pi a/W)}$	$Y \approx 1.12$ $(a/W \to 0)$	$Y \approx 1.12$ $(a/W \to 0)$	$K = 2\sigma\sqrt{(a/\pi)}$

Fig 4.2 Common cracked geometries

Geometry	Y or K expression
Centre crack 2a, finite width W (Feddersen)	$Y = \sqrt{\sec(\pi a/W)}$ $(a/W < \sim 0.7)$
Single edge crack (SENT, $a/W \le 0.6$)	$Y = 1.12 - 0.231(a/W) + 10.55(a/W)^2 - 21.72(a/W)^3 + 30.39(a/W)^4$; small a: **1.12** (free-surface factor)
Double edge cracks, small a/W	$Y \approx 1.12$
Embedded penny-shaped crack, radius a	$K = 2\sigma\sqrt{(a/\pi)} = (2/\pi)\,\sigma\sqrt{(\pi a)} = 0.637\,\sigma\sqrt{(\pi a)}$
Embedded elliptical crack ($a < c$)	$K = \sigma\sqrt{(\pi a/Q)}\,[\sin^2\varphi + (a^2/c^2)\cos^2\varphi]^{1/4}$, $Q \approx 1 + 1.464\,(a/c)^{1.65}$
Semi-elliptical surface crack (deepest point)	$K \approx 1.12\,\sigma\,\sqrt{(\pi a/Q)}$
Thin cylinder, long. through crack (half-length a)	$K = (pR/t)\,\sqrt{(\pi a)} \cdot M$, $M \approx \sqrt{(1 + 1.61\,a^2/(Rt))}$ (Folias bulging)
Cracks at a hole (short, symmetric)	$K \approx 3.39\,\sigma\,\sqrt{(\pi a)}$ $(a \ll r)$; for $a \gg r$ treat as crack $2(a + r)$
Pressure p in crack / wedge force P at centre	$K = p\sqrt{(\pi a)}$; $K = P/\sqrt{(\pi a)}$ (P per unit thickness)

3.1 Standard test specimens (ASTM E399) – $\alpha = a/W$

$$CT: K_I = \frac{P}{B\sqrt{W}} \cdot f(\alpha)$$

$f(\alpha) = (2+\alpha)(0.886 + 4.64\alpha - 13.32\alpha^2 + 14.72\alpha^3 - 5.6\alpha^4) \,/\, (1-\alpha)^{3/2}$

$$SENB\ (3\text{-pt},\ S = 4W): K_I = \frac{P\,S}{B\,W^{3/2}} \cdot f(\alpha)$$

$f(\alpha) = 3\sqrt{\alpha}\,[1.99 - \alpha(1-\alpha)(2.15 - 3.93\alpha + 2.7\alpha^2)] \,/\, [2(1+2\alpha)(1-\alpha)^{3/2}]$

- At $\alpha = 0.5$: $f_{CT} = 9.66$, $f_{SENB} = 2.66$.
- **Superposition:** K for combined loads = ΣK_i (same mode). Residual stress, thermal stress and pressure can each be handled separately. **Weight functions** give K for any stress distribution along crack line.
- Sources: Tada–Paris–Irwin handbook, Rooke & Cartwright, Murakami, Sih; FE (quarter-point elements, J-integral).

Worked Example 2 (SENT)

Strip $W = 50$ mm, edge crack $a = 10$ mm ($a/W = 0.2$), $\sigma = 200$ MPa. $Y = 1.12 - 0.0462 + 0.4220 - 0.1738 + 0.0486 = 1.371 \Rightarrow K = 1.371 \times 200 \times \sqrt{(\pi \times 0.01)} =$ **48.6 MPa√m**.

Worked Example 3 (CCT)

Plate $W = 100$ mm, $2a = 40$ mm, $\sigma = 150$ MPa. $\pi a/W = \pi \times 0.02/0.1 = 0.628$ rad $\Rightarrow \sec = 1.236$, $Y = \sqrt{1.236} = 1.112 \Rightarrow K = 1.112 \times 150 \times \sqrt{(\pi \times 0.02)} =$ **41.8 MPa√m**. (Infinite plate would give 37.6: finite width raises K by 11 %.)

Worked Example 4 (CT specimen)

$W = 50$ mm, $B = 25$ mm, $a = 25$ mm ($\alpha = 0.5$), $P = 20$ kN. $P/(B\sqrt{W}) = 20000/(0.025 \times 0.2236) = 3.58 \times 10^6$ Pa√m $\Rightarrow K = 3.58 \times 10^6 \times 9.66 =$ **34.6 MPa√m**.

Worked Example 5 (pressure vessel)

Thin cylinder $R = 500$ mm, $t = 10$ mm, $p = 5$ MPa $\Rightarrow \sigma_h = pR/t = 250$ MPa. Longitudinal through crack, half-length $a = 15$ mm: $M = \sqrt{(1 + 1.61 \times 0.015^2/(0.5 \times 0.01))} = 1.036 \Rightarrow K = 250\sqrt{(\pi \times 0.015)} \times 1.036 =$ **56.2 MPa√m**. If $K_{Ic} = 60$ MPa√m, margin is slim ⇒ **leak-before-break** check needed.

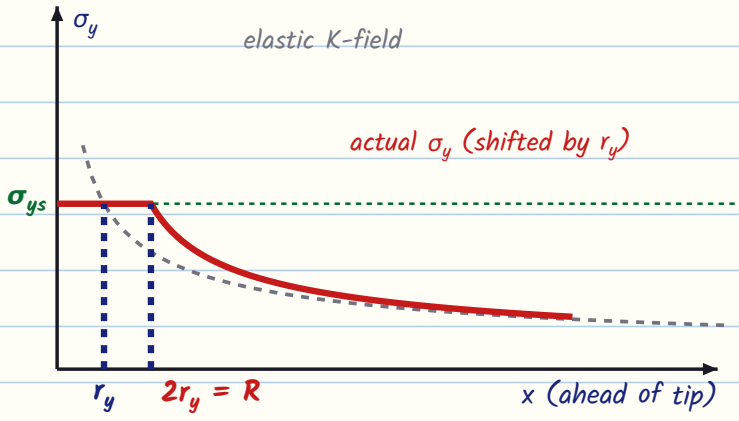
Worked Example 6 (surface crack)

Semi-elliptical surface crack $a = 5$ mm, $2c = 20$ mm ($a/c = 0.5$), $\sigma = 150$ MPa. $Q = 1 + 1.464(0.5)^{1.65} = 1.466 \Rightarrow K = 1.12 \times 150 \times \sqrt{(\pi \times 0.005/1.466)} =$ **17.4 MPa√m** at the deepest point.

4. Modelling Plastic Deformation at the Crack Tip

Elastic solution gives $\sigma_y \to \infty$ at tip, but real metals yield at $\sigma_{ys} \Rightarrow$ a **plastic zone** forms. If its size $r_p \ll a$ (small-scale yielding) LEFM remains valid with a small correction.

4.1 Irwin's model (1960)



Irwin's argument

$\sigma_y = K/\sqrt{(2\pi r)} = \sigma_{ys}$ at $r = r_y$

Plane stress:

$$r_y = (1/2\pi)(K/\sigma_{ys})^2$$

Load carried by cut-off area is redistributed ⇒ true zone = $2r_y$

Effective crack: $a_{eff} = a + r_y$

Fig 4.3 Irwin's plastic-zone correction

1. Elastic stress on crack line $\sigma_y = K/\sqrt{(2\pi r)}$. Yielding when $\sigma_y = \sigma_{ys} \Rightarrow r = r_y = (1/2\pi)(K/\sigma_{ys})^2$ (first estimate; plane stress).

2. In the yielded region σ_y cannot exceed σ_{ys}; the excess load (area under elastic curve above σ_{ys}) must be carried farther ahead ⇒ the zone doubles: **plastic zone size $2r_y = (1/\pi)(K/\sigma_{ys})^2$** (the equilibrium argument: $\sigma_{ys}\cdot R = \int_0^{r-y} K/\sqrt{(2\pi r)}\,dr \Rightarrow R = 2r_y$).
3. The plastic zone makes the crack behave as if it were **longer**: effective crack length **$a_{eff} = a + r_y$** (tip moves to the middle of the zone).

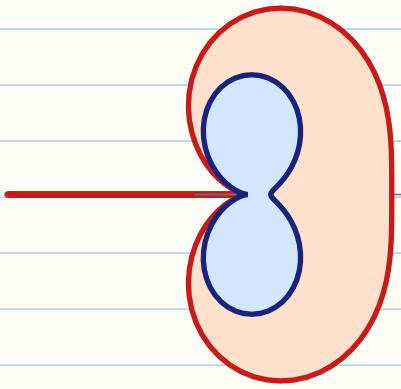
$$K_{eff} = Y\,\sigma\,\sqrt{(\pi\,(a + r_y))}$$

iterate; for CCT: $K_{eff} = K/\sqrt{(1 - \tfrac{1}{2}(\sigma/\sigma_{ys})^2)}$

Irwin plastic-zone formulae

> Plane stress: $r_y = (1/2\pi)(K/\sigma_{ys})^2$; zone size $2r_y = (1/\pi)(K/\sigma_{ys})^2$.
> Plane strain (σ_y constrained to ≈ $\sqrt{3}\,\sigma_{ys}$): $r_y = (1/6\pi)(K/\sigma_{ys})^2$ ≈ 1/3 of plane stress.
> CTOD (Irwin): $\delta_t = 2v(r_y) = (4/\pi)\,K^2/(E\,\sigma_{ys})$.
> Thickness: zone ≈ B ⇒ plane stress; $B \geq 2.5(K_{Ic}/\sigma_{ys})^2$ ⇒ plane strain.

4.2 Shape of the plastic zone



plane stress (large, 'dog-bone'/ round)

plane strain (small, 'butterfly' lobes)

von Mises, $S = K^2/4\pi\sigma_{ys}^2$

σ-state: $r_p(\theta) = S[1 + 1.5\sin^2\theta + \cos\theta]$

ε-state: $r_p(\theta) = S[1.5\sin^2\theta + (1-2\nu)^2(1+\cos\theta$

Fig 4.4 von Mises plastic zones – plane stress vs plane strain

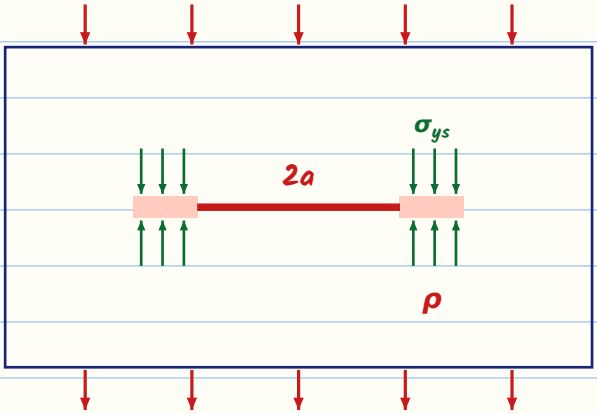
- Tresca & von Mises criteria applied to the K-field give zone shapes: plane stress – large, extends mostly **ahead & at ≈ 70°**; plane strain – small, **butterfly** lobes tilted back at 70°, almost no extent ahead ($\sigma_3 = 2\nu\sigma$ reduces shear).
- Hardening materials: smaller zone; large zone at high σ/σ_{ys} (>0.5) ⇒ need EPFM.
- Cyclic loading: reversed (cyclic) plastic zone $r_c \approx (1/2\pi)(\Delta K/2\sigma_{ys})^2$ = **¼ of the monotonic zone at ΔK**; controls closure & retardation (Unit V).

Worked Example 7

Plate, σ = 200 MPa, $2a$ = 40 mm, σ_{ys} = 400 MPa. $K = 200\sqrt{(\pi\times0.02)} = 50.1$ MPa√m.

r_y(stress) $= (1/2\pi)(50.1/400)^2 =$ **2.5 mm** (plane strain 0.83 mm). $a_{eff} = 22.5$ mm ⇒ K_{eff} ≈ **53.6 MPa√m** (7 % higher); CTOD (Irwin) $= (4/\pi)K^2/(E\sigma_{ys}) =$ **0.038 mm**.

4.3 Dugdale strip-yield model (1960)



Dugdale (strip-yield) model:

thin plastic strip ρ at each tip,
closing stress = σ_{ys} (perfectly plastic)

Real crack 2a → fictitious 2(a+ρ)

No stress singularity: $K_\sigma + K_{closure} = 0$

$\rho = a\,[\sec(\pi\sigma/2\sigma_{ys}) - 1]$

$\delta = (8\sigma_{ys}\,a/\pi E)\ln\sec(\pi\sigma/2\sigma_{ys})$

Fig 4.5 Dugdale model – thin plane-stress sheets

Assumptions: thin sheet (plane stress), **elastic-perfectly plastic**, plasticity confined to a thin strip of length ρ ahead of each tip, strip carries uniform stress σ_{ys}, crack behaves as one of length $2c = 2(a + \rho)$.

1. Problem (a): crack 2c under remote $\sigma \Rightarrow K_\sigma = \sigma\sqrt{(\pi c)}$.
2. Problem (b): crack 2c with closing stress σ_{ys} on the strips $a < |x| < c \Rightarrow K_{closure} = -2\sigma_{ys}\sqrt{(c/\pi)}\cos^{-1}(a/c)$.
3. Stress finite at the tip $\Rightarrow K_\sigma + K_{closure} = 0 \Rightarrow \sigma\sqrt{(\pi c)} = 2\sigma_{ys}\sqrt{(c/\pi)}\cos^{-1}(a/c)$.

$$a/c = \cos(\pi\sigma / 2\sigma_{ys}) \Rightarrow \rho = a\,[\sec(\pi\sigma/2\sigma_{ys}) - 1]$$

$$\delta_t = (8\sigma_{ys}\,a/\pi E)\ln\sec(\pi\sigma/2\sigma_{ys}) \quad \text{CTOD}$$

- **Small σ/σ_{ys}:** $\rho \approx (\pi^2/8)\,a\,(\sigma/\sigma_{ys})^2 = (\pi/8)(K/\sigma_{ys})^2$; $\delta_t \approx K^2/(E\,\sigma_{ys}) = G/\sigma_{ys}$.
- As $\sigma \to \sigma_{ys}$: $\rho \to \infty$ and $\delta \to \infty$ (plastic collapse). Critical stress from CTOD criterion $\delta = \delta_c$: $\sigma_c = (2\sigma_{ys}/\pi)\cos^{-1}[\exp(-\pi E\delta_c/8\sigma_{ys}\,a)]$.

Aspect	Irwin	Dugdale
Zone ($\sigma << \sigma_{ys}$)	$2r_y = K^2/\pi\sigma_{ys}^2 = 0.318\,K^2/\sigma_{ys}^2$	$\rho = (\pi/8)K^2/\sigma_{ys}^2 = 0.393\,K^2/\sigma_{ys}^2$
CTOD ($\sigma << \sigma_{ys}$)	$(4/\pi)\,K^2/E\sigma_{ys} = 1.27\,K^2/E\sigma_{ys}$	$K^2/(E\sigma_{ys})$
Material	elastic + yield cut-off	elastic-perfectly plastic strip
Valid range	$\sigma/\sigma_{ys} < \sim 0.5$	all $\sigma/\sigma_{ys} < 1$ (thin sheets)

Worked Example 8 (Dugdale)

$\sigma/\sigma_{ys} = 0.5 \Rightarrow \rho = a(\sec 45° - 1) = \mathbf{0.414\,a}$; $\sigma/\sigma_{ys} = 0.8 \Rightarrow \rho = a(\sec 72° - 1) = \mathbf{2.24\,a}$ (zone larger than the crack!).

Critical stress: E = 210 GPa, σ_{ys} = 400 MPa, a = 25 mm, δ_c = 0.1 mm: exponent = $\pi\times210\times10^9\times10^{-4}/(8\times400\times10^6\times0.025) = 0.825 \Rightarrow \exp(-0.825) = 0.438 \Rightarrow \cos^{-1} = 1.117$ rad ⇒

$\sigma_c = (2\times400/\pi)\times1.117 =$ **284 MPa** ($0.71\ \sigma_{ys}$).

4A. More on Multi-parameter Fields and SIFs

A. How the Williams series arises (outline)

1. Take $\varphi = r^{\lambda+1} F(\theta)$. $\nabla^4\varphi = 0 \Rightarrow F = A\cos(\lambda+1)\theta + B\cos(\lambda-1)\theta + C\sin(\lambda+1)\theta + D\sin(\lambda-1)\theta$.
2. Symmetry (Mode I) keeps cosine terms; flank conditions $\sigma_{\theta\theta} = \tau_{r\theta} = 0$ at $\theta = \pm\pi$ give two homogeneous equations in A, B.
3. Non-trivial solution needs determinant zero $\Rightarrow$ **$\sin(2\pi\lambda) = 0 \Rightarrow \lambda = n/2$** ($n = 1, 2, 3 \ldots$). Finite energy $\Rightarrow \lambda > 0 \Rightarrow$ lowest is $\lambda = 1/2$ (singular term).
4. Each λ gives one coefficient A_n fixed by the **global** geometry and loading; only A_1 (K) and A_2 (T) have a standard meaning.

B. Plastic-zone size table (plane stress / strain, Irwin)

K (MPa√m)	σ_{ys} (MPa)	r_y plane stress (mm)	r_y plane strain (mm)
30	400	0.90	0.30
50	400	2.49	0.83
80	400	6.37	2.12
50	800	0.62	0.21
100	800	2.49	0.83

- Same $K/\sigma_{ys} \Rightarrow$ same r_y: high-strength alloys have small zones $\Rightarrow$ LEFM works for them; low-strength steels need EPFM.
- **Validity check of LEFM:** require $r_y \ll a, (W-a), B$. Rule of thumb $a, B, (W-a) \geq 2.5(K/\sigma_{ys})^2$ ($\approx 16\ r_y$ in plane stress). For K = 80, σ_{ys} = 400: $(K/\sigma_{ys})^2 = 40$ mm² $\Rightarrow$ need $\geq$ 100 mm sections.

C. Superposition and weight functions

$$K_I = (1/\sqrt{(\pi a)}) \int_{-a}^{a} \sigma(x) \sqrt{((a + x)/(a - x))}\, dx$$

right tip, crack 2a

- $\sigma(x)$ = stress that would exist on the crack line **in the uncracked body**. Handles residual stress, thermal gradients, bending and pressure.

Example (edge crack, small a): membrane 80 MPa + bending (surface) 60 MPa, a = 4 mm: $K = (80 + 60)\times1.12\times\sqrt{(\pi\times0.004)} =$ **17.6 MPa√m**.

Example (weld residual stress): applied 150 MPa and tensile residual stress 200 MPa on an edge crack a = 5 mm: $K_{app} = 21.1$, $K_{res} = 28.1 \Rightarrow K_{total} =$ **49.1 MPa√m** – residual stress contributes more than the applied load! (relieve by PWHT or peening).

D. Choice of specimen and constraint

- Thick, deeply cracked bend/CT specimens: high constraint ($T \geq 0$) ⇒ conservative **K_{Ic}**.
- Thin plates, shallow/ surface cracks, centre-cracked panels: low constraint ⇒ higher apparent toughness; using K_{Ic} is conservative; **constraint-corrected toughness** (J–Q, K–T) can reduce over-conservatism in structural integrity assessment.

E. Dugdale CTOD numerical

$a = 20$ mm, $\sigma/\sigma_{ys} = 0.3$, $\sigma_{ys} = 400$ MPa, $E = 210$ GPa: $\delta = (8\times400\times10^{6}\times0.02/(\pi\times210\times10^{9}))$ $\ln\sec(0.15\pi) = 9.70\times10^{-5} \times 0.1154 =$ **0.0112 mm** (Irwin small-scale: $(4/\pi)K^2/E\sigma_{ys} \approx 0.0137$ mm).

F. Short questions

Q. What does a negative T-stress do? Reduces crack-tip triaxiality (loss of constraint) ⇒ larger plastic zone, higher fracture toughness, stable crack path.

Q. Why does the plastic zone make a crack effectively longer? Yielded material cannot carry the elastic stress; the stress field is displaced forward by r_y, equivalent to a crack of length $a + r_y$.

Q. When is Dugdale preferred to Irwin? Thin sheets in plane stress (steel sheets) with high $\sigma/\sigma_{ys} > 0.5$, where the zone is a narrow strip.

Q. State two uses of CTOD. Fracture criterion for ductile steels (welds, BS 7448), measure of toughness when LEFM is invalid.

Q. Which mode has K_c lowest in most metals? Mode I (plane strain): $K_{Ic} < K_{IIc} < K_{IIIc}$.

G. Practice problems (with answers)

1. Centre crack $2a = 30$ mm, $W = 90$ mm, $\sigma = 120$ MPa. Find K_I. Ans: $\pi a/W = 0.524$, sec = 1.155, $Y = 1.075$ ⇒ $K = 1.075\times120\times\sqrt{(\pi\times0.015)}$ = **28.0 MPa√m**.
2. Find r_y (plane stress & strain) for $K = 60$ MPa√m, $\sigma_{ys} = 500$ MPa. Ans: $(60/500)^2/(2\pi)$ = **2.29 mm**; plane strain **0.76 mm**.
3. Inclined crack $\beta = 45°$, $\sigma = 100$ MPa, $a = 10$ mm. Ans: $K_I = K_{II} = 0.5\times17.72$ = **8.86 MPa√m**; $G = 2\times8.86^2/E'$.
4. Penny-shaped crack radius 3 mm in $\sigma = 200$ MPa. Ans: $K = 2\times200\times\sqrt{(0.003/\pi)}$ = **12.4 MPa√m**.
5. Using Dugdale, find ρ/a for $\sigma/\sigma_{ys} = 0.3$. Ans: $\sec(27°) - 1$ = **0.122**.

4B. Mixed-mode Details

A. General biaxial loading on an inclined crack

$$K_I = (\sigma_y \cos^2\beta + \sigma_x \sin^2\beta)\sqrt{(\pi a)}$$

$$K_{II} = (\sigma_y - \sigma_x)\sin\beta\cos\beta\sqrt{(\pi a)}$$

- β measured from the plane $\perp$ to σ_y. For **equi-biaxial** loading ($\sigma_x = \sigma_y$) $K_{II} = 0$ and $K_I = \sigma\sqrt{(\pi a)}$ for any crack orientation – no mixed mode.

Example: $\sigma_y = 100$, $\sigma_x = 50$ MPa, $\beta = 30°$, $a = 10$ mm: $K_I = (75 + 12.5)\times0.1772 =$ **15.5 MPa√m** ; $K_{II} = 50\times0.433\times0.1772 =$ **3.84 MPa√m**.

B. Where does mixed mode occur?

Situation	Reason
Inclined / misaligned weld defects	Crack not $\perp$ to principal stress
Shafts: torsion + bending	Helical (45°) cracks, Mode I + III
Rolling contact, gears, rails	Shear & compression: Mode II / I + II
Delamination of composites, adhesive joints	Interface cracks always mixed mode
Pressure vessels with nozzles, riveted lap joints	Bending + membrane + shear

C. Strain-energy-density factor (Sih) – preview

$$S = a_{11} K_I^2 + 2\, a_{12} K_I K_{II} + a_{22} K_{II}^2 + a_{33} K_{III}^2$$

- $a_{11} = (1/16\pi\mu)(1+\cos\theta)(\kappa - \cos\theta)$; $a_{12} = (1/16\pi\mu)\sin\theta\,[2\cos\theta - (\kappa-1)]$; $a_{22} = (1/16\pi\mu)[(\kappa+1)(1-\cos\theta) + (1+\cos\theta)(3\cos\theta - 1)]$; $a_{33} = 1/(4\pi\mu)$.
- Strain energy density near tip $dW/dV = S/r$. **Crack grows toward the minimum of $S(\theta)$**; fracture when $S_{min} = S_c$. For pure Mode I S_{min} is at $\theta = 0$.

D. Summary of near-tip fields on the crack line ($\theta = 0$)

	Mode I	Mode II	Mode III
Non-zero stress	$\sigma_x = \sigma_y = K_I/\sqrt{(2\pi r)}$	$\tau_{xy} = K_{II}/\sqrt{(2\pi r)}$	$\tau_{yz} = K_{III}/\sqrt{(2\pi r)}$
K for centre crack	$\sigma\sqrt{(\pi a)}$	$\tau\sqrt{(\pi a)}$	$\tau\sqrt{(\pi a)}$
Energy release G	K_I^2/E'	K_{II}^2/E'	$K_{III}^2/2\mu$
Opening	normal (v)	sliding (u)	tearing (w)

E. Crack-tip blunting – physical picture of CTOD

- A real crack tip blunts under load: tip radius grows from $\rho \approx 0$ to $\rho \approx \delta/2$. The opening at the original tip position δ_t is a measure of **local strain** at the tip.
- In SSY, CTOD ↔ K ↔ J ↔ G are interchangeable: $\delta_t = K^2/(m\,\sigma_{ys}\,E') = G/(m\,\sigma_{ys}) = J/(m\,\sigma_{ys})$, with $m \approx 1$ (plane stress) to 2 (plane strain) (Dugdale $m = 1$).
- CTOD is the fracture parameter of choice for **structural steels / welds** (BS 7448, ISO 12135): fracture when $\delta = \delta_c$.

4C. Irwin's Derivation in Detail & Typical Toughness Values

A. Equilibrium argument for the zone size $2r_y$

1. Elastic stress on the crack line: $\sigma_y = K/\sqrt{(2\pi r)}$. It equals σ_{ys} at $r = r_y = K^2/(2\pi\sigma_{ys}^2)$.
2. Truncating σ_y at σ_{ys} removes load: area under the elastic curve for $0 < r < r_y$ is $\int K/\sqrt{(2\pi r)}dr = 2K\sqrt{(r_y/2\pi)} = 2\sigma_{ys}\,r_y$, while the truncated stress carries only $\sigma_{ys}\,r_y$.
3. The missing force $\sigma_{ys}\,r_y$ must be carried ahead. To keep equilibrium the zone extends to R with $\sigma_{ys}\,R = 2\sigma_{ys}\,r_y \Rightarrow$ **$R = 2r_y$**.
4. The stress curve outside the zone is the elastic one shifted by r_y: effective tip at $a + r_y$, so $K_{eff} = Y\sigma\sqrt{(\pi(a + r_y))}$.
5. Plane strain: constraint raises σ_y at yield to $\approx \sqrt{3}\sigma_{ys}$ (Tresca/von Mises, $\sigma_z = \nu(\sigma_x + \sigma_y)$) ⇒ r_y reduces by 3.

B. Typical fracture toughness K_{Ic} (room temperature)

Material	σ_{ys} (MPa)	K_{Ic} (MPa√m)
Aluminium 7075-T6	500	≈ 29
Aluminium 2024-T3	350	≈ 34
Titanium Ti-6Al-4V	900	≈ 55 – 90
High-strength steel 4340 (tempered low)	1500	≈ 50
Mild / structural steel (above DBTT)	250	> 100 (EPFM required)
Alumina, Si_{3N4} ceramics	—	≈ 3 – 6
Glass / PMMA	—	≈ 0.7 / 1.0 – 1.5
Concrete	—	≈ 0.2 – 1.4

- Trade-off: higher σ_{ys} → lower K_{Ic}. Critical size for 7075-T6 at σ = 300 MPa ($Y = 1$): $a_c = (1/\pi)(29/300)^2 =$ **3.0 mm** — very small ⇒ NDT must detect millimetre cracks; mild steel tolerates much larger flaws.
- Plane-zone check for 7075-T6 at fracture: $r_y(\text{strain}) = (1/6\pi)(29/500)^2 =$ **0.18 mm**,

$<< B$ and $a \Rightarrow$ LEFM valid.

5. Exam Pointers (Unit IV)

1. Explain the Williams series and the T-stress. How does T affect constraint and fracture toughness?
2. Write the near-tip stress field for Modes II and III. Derive K_I, K_{II} for an inclined crack.
3. Evaluate K for SENT / CT / penny-shaped / surface crack (numerical).
4. Derive Irwin's plastic-zone size (plane stress & strain) and the effective crack length. Compute K_{eff}.
5. Derive Dugdale's plastic strip length and CTOD; compare with Irwin.

Formula sheet – Unit IV

> $K = Y\sigma\sqrt{(\pi a)}$; CCT $Y = \sqrt{\sec(\pi a/W)}$; edge $Y = 1.12$; penny $K = 2\sigma\sqrt{(a/\pi)}$

> $r_y = (1/2\pi)(K/\sigma_{ys})^2$ (σ) ; $(1/6\pi)(K/\sigma_{ys})^2$ (ε) ; $a_{eff} = a + r_y$

> Dugdale $\rho = a[\sec(\pi\sigma/2\sigma_{ys}) - 1]$; $\delta = (8\sigma_{ys}\, a/\pi E) \ln \sec(\pi\sigma/2\sigma_{ys}) \rightarrow K^2/E\sigma_{ys}$

> Inclined crack: $K_I = \sigma\sqrt{(\pi a)}\cos^2\beta$; $K_{II} = \sigma\sqrt{(\pi a)}\, \sin\beta\, \cos\beta$